Haotian Yin

PhD Candidate in Computer Science

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Research Interests

Proxy re-encryption • Lattice-based cryptography • Provable security

Education

2023–present	PhD Candidate in Computer Science, Xi'an Jiaotong-Liverpool University (XJTLU)
2020–2023	Master of Science, Taiyuan University of Technology (TYUT)
2013–2017	Bachelor of Management, Chongqing University (CQU)

Current Research Manuscripts

Lattice-Based Proxy Re-Encryption: Reconstruction Attacks and HRA-Secure Constructions. **Haotian Yin**, Jie Zhang, Yuji Dong, Dominik Wojtczak, and Eng Gee Lim. *IACR Communications in Cryptology (CiC) – Under Review; IACR Cryptology ePrint Archive, Paper 2025/704, revised 2026.*

Single-Hop HRA-Secure Owner-Encrypted Lattice-Based Proxy Re-Encryption without Statistical Noise Flooding. **Haotian Yin**, Jie Zhang, Yuji Dong, Dominik Wojtczak, and Eng Gee Lim. *Designs, Codes and Cryptography – Major Revision; revision in progress; IACR Cryptology ePrint Archive, Paper 2026/1113.*

Selected Peer-Reviewed Publications

Collusion-Safe Proxy Re-Encryption. **Haotian Yin**, Jie Zhang, Wanxin Li, Yuji Dong, Eng Gee Lim, and Dominik Wojtczak. *Accepted at Financial Cryptography and Data Security 2026 (FC 2026); proceedings forthcoming.*

Updatable Signature with Public Tokens. **Haotian Yin**, Jie Zhang, Wanxin Li, Yuji Dong, Eng Gee Lim, and Dominik Wojtczak. *Journal of Information Security and Applications, 91, 2025, Article 104058.*

LSNCP: Lightweight and Secure Numeric Comparison Protocol for Wireless Body Area Networks. **Haotian Yin**, Xin Huang, Bin Xing, Jiajia Huang, Xiaoxin Sun, Jianshuang Li, Sheng Chai, Di Zhang, Rana Abu Bakar, and Wei Wang. *IEEE Internet of Things Journal, 10(15), 2023, pp. 13247–13263.*

Security Analysis of Bluetooth Secure Simple Pairing Protocols with Extended Threat Model. **Haotian Yin**. *Journal of Information Security and Applications, 72, 2023, Article 103385.*

Talks and Presentation

9 Sep 2026	On the Security and Construction of Lattice-Based Proxy Re-Encryption Invited seminar talk, PQC-X Lab, Suzhou, China.
2026	Collusion-Safe Proxy Re-Encryption Conference presentation, FC 2026, St. Kitts and Nevis.

Research Experience

Oct–Nov 2026	Research Assistant, PQC-X Lab Research on NTRU attacks and CBOM detection.
2026	Visiting PhD Researcher, University of Liverpool Host: Prof. Dominik Wojtczak. Research on HRA and CCA security notions for proxy re-encryption.
2021–2023	Blockchain-Based IoT Edge Computing Model and Its Core Mechanism Applied Basic Research in Shanxi Province. Research on lightweight authentication protocols and their security analysis.
2020	Internet of Things Computing Model and Its Basic Mechanism Based on Blockchain National Natural Science Foundation of China (NSFC). Research on secure device-pairing protocols.

Honors and Scholarships

2025	2024 Web3.0 Innovation Competition, Excellence Award
2023	TYUT Postgraduate Scholarship, Special Class
2021	Outstanding Postgraduate of TYUT

Languages

Chinese (native) • English (IELTS 7)